

AMOGH SOMISETTY

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EE junior building GPU cluster infrastructure: shipped an open-source GPU fleet-reliability agent (PyPI, Docker, CI, Prometheus) that runs as a daemon on production clusters and was blind-validated on 72 H100s. Seeking a Fall 2026 GPU / cloud-infrastructure internship.

EDUCATION

California Polytechnic State University, San Luis Obispo

B.S. Electrical Engineering

Expected June 2028

San Luis Obispo, CA

- **Coursework:** Computer Architecture, Digital Logic Design, Embedded Systems, Circuits I-III, Signals & Systems

EXPERIENCE

Founder & Lead Researcher

Theta — open-source GPU fleet-reliability agent (ThermalOS research)

May 2026 – Present

San Luis Obispo, CA

- Built and shipped **Theta** (`pip install runtheta`): an agent that packages, validates, and distributes GPU health monitoring, released on PyPI with Docker images, CI across Python 3.10–3.12, and a one-command reproduce script.
- Runs as a long-lived daemon on GPU clusters (CLI → daemon architecture with a **theta setup** wizard); exports **Prometheus** metrics and webhook alerts, and consumes NVML/DCGM telemetry from SLURM-scheduled production nodes.
- Deployed as a **Kubernetes DaemonSet** via a self-authored **Helm** chart: per-node agents with a Prometheus-discovered metrics service, liveness/readiness probes, hardened non-root security context, and GPU-node scheduling (runtimeClass, node selectors, taint tolerations) aligned with the NVIDIA GPU Operator; verified end-to-end on a k3d cluster.
- Blind-flagged 3 degraded units across **72 H100 SXM5 GPUs** (Princeton Della cluster) from DCGM/NVML telemetry; top flag independently re-confirmed 3 months later by Princeton's own DCGM diagnostic at **+47.8%** thermal resistance, caught pre-throttle.
- Wrote a **Go** Kubernetes controller (**theta-node-labeler**, client-go) that reconciles **theta.dev/gpu-health** node labels from agent telemetry (observe → scrape → patch loop, minimal RBAC, distroless image); CI runs **helm lint**, **go vet**, and a full **k3d end-to-end test** asserting DaemonSet readiness, metrics scrape, and label convergence on every PR.
- Shipped **theta top**, a live fleet TUI with one-command Kubernetes attach (`--k8s: pod discovery, auto port-forward, cleanup`) plus a 6-layer detection stack; preparing a publication targeting ICPE 2027.

Edge-AI Hardware Designer — Carrier Boards & Sensor Modules

Poly UAS (Cal Poly Unmanned Aerial Systems)

Apr 2026 – Present

San Luis Obispo, CA

- Design the carrier board and sensor modules for the team's **NVIDIA Jetson AGX Orin** edge-AI stack: power delivery, sensor interfacing, and size/weight/power budgeting for onboard real-time computer vision.

Lab Operations Intern: Systems, Automation & Validation

Natera

June 2025 – Sept 2025

Pleasanton, CA

- Built Python automation for barcode validation and pathology-report retrieval, removing manual lookups across the team.
- Cut validation and packaging cycle time from ~20 min to 10–12 min (>40%) by mapping the workflow and eliminating redundant verification steps in a high-throughput production line.

TECHNICAL PROJECTS

funck.live — Event-Ticketing Platform in Production | *React, TypeScript, Supabase, Stripe*

- Built and operate a live ticketing platform with real paying users: Stripe payments with server-side webhook verification, ~40 serverless edge functions on Supabase (PostgreSQL), and role-based access control; sole developer from design through production operations.

RISC-V CPU (RV32I) on FPGA | *SystemVerilog, Vivado, Basys-3*

- Designed a multi-cycle RV32I CPU from the RTL up (full ISA, 10-operation ALU, FSM control unit, 32×32 register file); synthesized and verified on a Xilinx Basys-3 through waveform analysis and iterative timing fixes.

TECHNICAL SKILLS

Languages & Tools: Python, Go (client-go), TypeScript, C/C++, Bash, SystemVerilog, Git, Linux

Infrastructure & Cloud: Kubernetes (Helm charts, DaemonSets, controllers, RBAC, k3d), Docker, CI (GitHub Actions), Prometheus, FastAPI, SLURM, PyPI packaging & distribution, Supabase/PostgreSQL, webhook systems

GPU & Telemetry: NVML (pynvml), DCGM, nvidia-smi, GPU power/thermal characterization, anomaly detection, NVIDIA Jetson (AGX Orin)

ML & Data: scikit-learn, pandas, NumPy, ensemble classifiers, Isolation Forest, survival analysis (lifelines)

LEADERSHIP

President, Poly Engineering Consulting: lead client-facing engineering project teams

Sept 2024 – Present

Lab Assistant, Cal Poly EPIC: teach embedded systems to precollegiate students

Summer 2026